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Graph feature selection for enhancing radiomic stability and reproducibility across multiple institutions in head and neck cancer

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Radiomic biomarkers offer promise for precision oncology. However, their clinical utility is limited by variability from differing imaging protocols and the high dimensionality of radiomics data. Feature selection is key for better interpretability, accuracy, and efficiency, yet traditional methods lack stability and reproducibility. We investigate a Graph-Based Feature Selection (Graph-FS) approach that models feature interdependencies to identify stable radiomic signatures for head and neck squamous cell carcinoma (HNSCC) across institutions. We retrospectively analyzed 1,648 radiomic features extracted from the gross tumor volumes of 752 HNSCC patients from three institutions. After standard preprocessing and applying 36 radiomics parameter configurations to simulate variability, we compared Graph-FS with established methods: Boruta, Lasso, Recursive Feature Elimination (RFE), and Minimum Redundancy Maximum Relevance (mRMR). We evaluated feature selection stability and reproducibility using Pearson correlation, the Jaccard Index (JI), and the Dice-Sorensen Index (DSI) and assessed ranking consistency with Kendall's Coefficient of Concordance (W). Graph-FS achieved higher stability (JI = 0.46, DSI = 0.62, OP = 45.8%) versus baseline methods with JI of 0.005 (Boruta), 0.010 (Lasso), 0.006 (RFE) and 0.014 (mRMR). These results demonstrate that Graph-FS enhances feature stability, reproducibility, and predictive performance. This method could facilitate integration into AI-driven radiomics workflows for reliable, multi-center biomarker discovery.

Radiomics holds significant promise in precision oncology by extracting high-dimensional, quantitative imaging features that reflect tumor phenotypes and predict clinical outcomes such as survival and treatment response[1,2]. However, a substantial challenge is the lack of reproducibility across different imaging protocols, scanners, and institutions, which limits the clinical translation of radiomic biomarkers[3–6].

Radiomic features are prone to variations in acquisition parameters, voxel resolution, normalization techniques, and filtering algorithms[6,7]. Although harmonization methods like ComBat can mitigate inter-site variability by adjusting for batch effects[8,9] they often require access to raw imaging data[10]. They may also be challenging to scale in multi-institutional settings. Moreover, the high dimensionality and collinearity of radiomic features increase the risk of overfitting, making robust feature selection essential for building generalizable predictive models[11,12].

Traditional feature selection methods such as Lasso[13] Boruta[14] Minimum Redundancy Maximum Relevance (mRMR)[15] and Recursive Feature Elimination (RFE)[16] have been widely used to reduce redundancy and enhance

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model robustness[12]. However, these methods often yield unstable feature sets when applied across different preprocessing settings or cross-validation splits, compromising reproducibility. Additionally, conventional approaches treat features as independent, ignoring their complex interdependence.

Graph-based methods offer a promising alternative by representing features as nodes in a network with edges that encode pairwise statistical dependencies. Although these approaches have been applied in brain connectomics, digital pathology, and multimodal data fusion[17] their use in radiomics, particularly for feature selection, remains limited. Most graph neural network (GNN)-based studies in radiomics focus on patient-level classification; for example, GCNs have been used to predict neurological conditions using imaging and demographic data[17] and other studies have combined radiomics with GNNs for pathology classification[18] or PET/CT-based risk stratification in head and neck cancer[19]. However, no one has addressed the critical challenge of selecting stable and reproducible features across heterogeneous imaging conditions and institutions.

To address this gap, we propose a Graph-Based Feature Selection (Graph-FS) method tailored for radiomics. Graph-FS constructs a feature similarity graph where each node represents a radiomic feature and edges represent statistical similarities (e.g., Pearson correlation). Features are grouped into connected components, and the most representative nodes are selected using centrality measures such as betweenness centrality. This approach preserves informative features by linking isolated nodes to their most similar neighbors, thus maintaining overall graph connectivity.

We evaluated Graph-FS using a multi-institutional cohort of 752 patients with squamous head and neck cell carcinoma (HNSCC) from three independent institutions. To simulate real-world variability, we systematically varied 36 radiomics parameter configurations. We compared Graph-FS with conventional selection methods (Lasso, Boruta, mRMR, and RFE) by assessing feature stability, ranking consistency, and cross-center reproducibility. Finally, we demonstrated the clinical utility of the selected features by predicting two-year survival using five machine-learning classifiers: Random Forest, CatBoost, XGBoost, AdaBoost, and LightGBM.

This study is the first to apply graph-based feature selection in radiomics, focusing specifically on enhancing reproducibility and robustness across diverse imaging conditions.

Key points

- The study utilizes a diverse dataset of 752 HNSCC cases from three leading institutions, which enhances generalizability and minimizes biases inherent to single-center studies. Validation across heterogeneous populations and imaging protocols reinforces the clinical applicability of Graph-FS.
- Graph-FS is introduced as a novel graph-based feature selection framework for radiomics, with the Connected Components variant achieving the highest improvements in feature stability and reproducibility.
- Our analysis reveals that feature importance rankings are highly sensitive to parameter variations, highlighting the limitations of traditional feature selection approaches and the risks of over-relying on top-ranked features in predictive modeling.
- We provide GFSIR, an open-source Graph-Based Feature Selection package for radiomics in Python, which integrates seamlessly into existing radiomics pipelines and supports broader adoption in research and clinical applications.

Methods

This study was approved by the University of Maryland School of Medicine Institutional Review Board (IRB) by the ethical standards of the IRB and the Declaration of Helsinki. A formal waiver of informed consent was granted by the University of Maryland School of Medicine IRB, as the study involved retrospective data collection that did not require individual patient consent. All patient data were fully anonymized to ensure confidentiality and were handled in compliance with the Health Insurance Portability and Accountability Act (HIPAA) regulations applicable to U.S.-based research. The study was registered under Clinical Trials Registry Number HP-00101754.

Study design and data collection

Figure 1 illustrates the overall workflow of this study. This retrospective analysis includes 752 patients with primary head and neck squamous cell carcinoma (HNSCC) treated across three medical centers.

The UMB cohort consists of 203 patients treated between 2006 and 2017, of whom 65% are male. Tumor locations are primarily in the oral cavity (82%), with the remaining cases involving the oropharynx and larynx. Postoperative treatment varied: 54% of patients received radiation therapy alone or chemotherapy, while 46% had surgery as their sole treatment[21].

The MD Anderson Cancer Center cohort includes 412 patients treated between 2003 and 2013, with data publicly accessible via the Cancer Imaging Archive (TCIA)[22]. In this cohort, 86% of patients are male, with most tumors located in the oropharynx, primarily at the base of the tongue (51%) and tonsils (40%). Concurrent chemotherapy was administered to 59% of patients, with 98% receiving platinum-based therapy. Additionally, 13% received postoperative radiation, with an average dose of 68.66 Gy over 28–40 daily fractions. Detailed patient and treatment characteristics are available on the TCIA website.

The UMMC dataset consists of 137 patients, predominantly male (81.5%), with 64% having tumors originating in the oropharynx. At the time of data collection, 107 patients (78%) were alive, while 30 (22%) were deceased. Treatment varied, with 74.1% receiving radiation therapy alone and 25.9% receiving combined chemo-radiation post-surgery. This dataset, which includes CT scans, manual tumor delineations, clinical data, and survival outcomes, was used to validate the prognostic value of the radiomic features[2,23].

Demographic and clinical information for all cohorts is summarized in Table 1. Differences in demographic and clinical characteristics across the three cohorts were evaluated using p-values for each variable. Welch's ANOVA was applied to compare mean ages, accounting for differences in sample sizes and variances. For

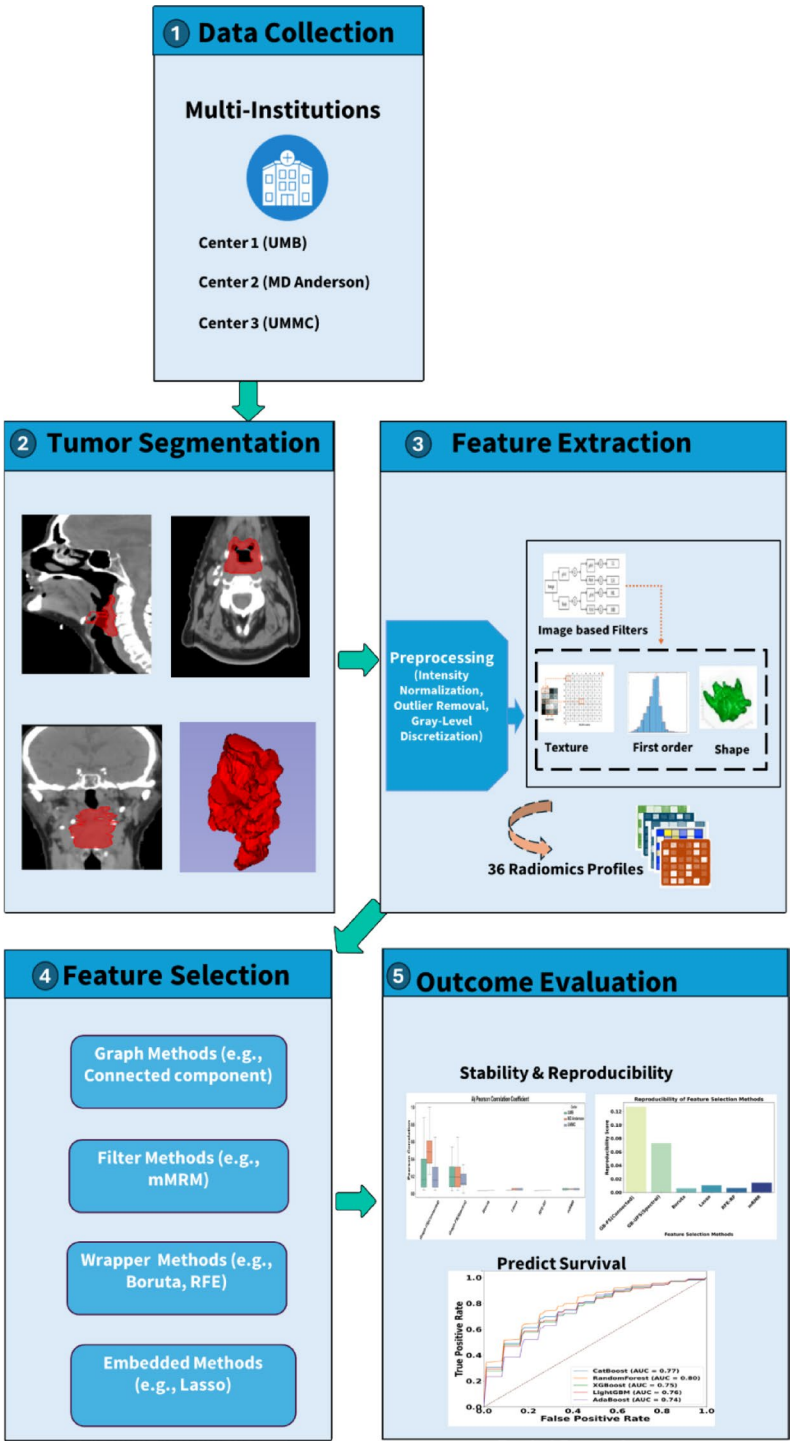

Fig. 1. Radiomics analysis pipeline for head and neck cancer across multiple institutions. (1) Pre-treatment CT scans were collected from three centers: the University of Maryland, Baltimore (UMB), MD Anderson Cancer Center, and MAASTRO Medical Center (UMMC). (2) Tumors were segmented to identify the primary tumor region (GTVp). (3) Radiomic features were extracted using original and filtered images (e.g., wavelet, LoG), categorized into texture, first-order, and shape features. Preprocessing included normalization, outlier removal, and discretization, generating 36 radiomic profiles. (4) Feature selection was performed using graph-based, filter, wrapper, and embedded methods. (5) Selected features were assessed for stability and reproducibility and then used to train ensemble models for survival prediction. This figure was created by the authors using Microsoft PowerPoint version 2505 (Build 18827.20128 Click-to-Run). Graph-FS (Connected) = Graph-Based Feature Selection (Connected Components), Graph-FS (Spectral) = Graph-Based Feature Selection (Spectral Clustering), Lasso = Least Absolute Shrinkage and Selection Operator, mRMR = Minimum Redundancy Maximum Relevance, RFE-RF = Recursive Feature Elimination with Random Forest.

Study	MD Anderson	UMB	UMMC	*p*-value
Subjects	412	203	137	
Age (year)	58 (29–80)	61(28–97)	62(44–83)	1.45e−05
Gender (%)	Male: 86%	Male: 65.50%	Male 81.5%	5.56e−09
	Female:14%	Female:34.50%	Female: 18.5%	
Status of 2 year-survival	Live:377 (91.5%)	Live:113 (55.6%)	Live:107 (78%)	5.1e−48
	Decease:35 (8.5%)	Decease:90 (44.4%)	Decease:30 (22%)	
Smoking status	Yes:60.5%	Yes: 64%	NA	0.44
	No: 39.5%	No: 36%		
T-category				0.0158
T1	87 (21.12%)	56 (27.6%)	35 (25.54%)	2.33e−08
T2	170 (41.26%)	54 (26.6%)	32 (23.36%)	
T3	93 (22.57)	29 (14.28%)	24 (17.52%)	
T4	62(15.05%)	64(31.52%)	46 (33.58%)	
N-category				8.46e−37
N0	37 (8.98%)	103 (50.74%)	60 (43.79)	
N1	45 (10.92%)	31 (15.27%)	16 (11.68%)	
N2	318 (77.18%)	55 (27.09%)	58 (42.33%)	
N3	12 (2.92%)	14 (6.90%)	3 (2.20%)	

Table 1. Demographic and clinical characteristics of patients across multiple institutions. NA = Not Available, T-Category = Tumor Stage Category, N-Category = Nodal Stage Category, p-value: Statistical significance value for comparisons across institutions.

categorical variables, including gender, T-category, N-category, smoking status, and 2-year survival status, Chi-square tests were used to examine distributional differences among cohorts. A p-value of < 0.05 was considered statistically significant, indicating meaningful differences in these characteristics.

Tumor delineation and preprocessing

Experienced radiation oncologists manually delineated the gross tumor volumes for primary disease (GTVp). These structures were labeled following the American Association of Physicists in Medicine (AAPM) TG-263 guidelines and saved in the RT-STRUCT format. The delineated contours for Center B and Center C are publicly available in TCIA[22].

To improve feature reproducibility and maintain consistency across cases, CT images and GTV contours underwent careful preprocessing. CT images were resampled to an isotropic voxel size of 1 mm x 1 mm x 1 mm using B-spline interpolation. The GTV contours were converted into voxel-based segmentation masks aligned with the resampled CT grid.

Feature extraction

Radiomic features were extracted from the Gross Tumor Volume of primary tumors (GTVp) in pre-treatment original CT scans and eight distinct image transformations to capture a comprehensive range of tumor phenotypes. These transformations included a series of Laplacian-of-Gaussian (LoG) filters with sigma values of 1, 2, 3, 4, and 5 mm to emphasize fine structural details and wavelet filters utilizing various combinations of high- and low-pass filters (LLL, HLL, LHL, LLH, LHH, HHL, HHH) to decompose spatial frequency information. Additional transformation approaches such as square, square root, logarithmic, exponential, gradient, and 2D and 3D local binary patterns (LBP) were also applied to enrich the feature set. This comprehensive extraction process resulted in a total of 1,648 features per patient, providing a robust foundation for analyzing and characterizing tumor heterogeneity.

To thoroughly assess feature stability across different preprocessing conditions, we systematically varied several parameters: normalization scales (50 and 100), discretized gray levels (5, 10, 15, 20, 25, 30), and outlier removal thresholds (2, 3, 4). This comprehensive exploration of preprocessing configurations enabled a robust evaluation of feature stability under diverse conditions.

Radiomic feature extraction was performed using PyRadiomics[24] (v3.1.0), which follows the standardized definitions established by the Image Biomarker Standardization Initiative (IBSI) to ensure consistent and reproducible feature calculations. All extracted features were standardized to a mean of 0 and a variance of 1 to ensure equal contribution to subsequent analyses.

Detailed PyRadiomics parameter settings and a comprehensive list of radiomic feature classes and types are provided in Supplementary Tables S1 and S2, respectively.

Feature selection methods

In radiomics, feature engineering aims to remove redundant, irrelevant, or highly correlated features to enhance model generalizability and reduce variance. Redundancy arises when multiple features convey the same information, often resulting in collinearity, where predictor variables are highly correlated. Collinearity

complicates model interpretation, increases the standard errors of coefficient estimates, and can produce unreliable predictions. Therefore, addressing collinearity is crucial for building robust and interpretable models.

Initially, we applied Kendall's W to identify and retain the most consistent features. This process helped eliminate unstable features, reducing dimensionality while ensuring that the retained features were reliable.

For benchmarking, established feature selection methods including Boruta, Lasso (Least Absolute Shrinkage and Selection Operator), Minimum Redundancy Maximum Relevance (mRMR), and Recursive Feature Elimination (RFE) were applied and compared against our proposed Graph-Feature Selection (Graph-FS) method.

Graph-FS constructs a weighted graph where each feature X_i is represented as a node, and edges S_{ij} between nodes represent pairwise similarity scores, typically derived from metrics like Pearson correlation or mutual information: $S_{ij} = similarity(X_i,\ X_j)$. This similarity matrix underlies clustering algorithms such as Spectral Clustering and Dominant Set Clustering, which group interdependent features based on the graph's structure[25,26]. Unlike traditional clustering which requires a fixed number of clusters, dominant sets form dynamically based on the data's structure. This flexibility makes the approach particularly effective for high-dimensional data with complex inter-feature correlations. Graph-FS selects representative features from each cluster by ranking nodes based on centrality metrics[27]. For example, degree centrality, $C\left(X_j\right) = \sum_{X_k \in neighbors\left(X_j\right)} S_{jk}$ reflects a feature's direct connectivity. At the same time, eigenvector centrality captures both direct and indirect influence within the graph.

We built the graph by connecting two features if their absolute Pearson correlation was higher than a threshold calculated as:

$Threshold = \mu + n\sigma$, where μ is the average of all pairwise correlations, and σ is their standard deviation. Using a heuristic approach, we found that setting $n = 0.3$ provided a good balance between graph sparsity and feature connectivity. This adaptive thresholding strategy, previously applied in graph network studies[28,28,30]allows the cutoff to adjust automatically based on the statistical properties of each dataset.

We then applied two complementary strategies on this graph to perform feature selection: Connected Components Analysis[31,32] and Spectral Clustering[33].

1. *Connected Components Analysis* identifies groups of highly similar features based on the graph's structure. First, we built an undirected similarity graph h $G = (V, E)$, where vertices v_i represent features and edges e_{ij} connects two features if their similarity S_{ij} exceeds a threshold. The Similarity metric used was the Pearson correlation coefficient: $S_{ij} = \frac{Cov(X_i,\ X_j)}{\sigma_{X_i} \cdot \sigma_{X_j}}$. Where $Cov(X_i,\ X_j)$ is the covariance between features, and σ_x is the standard deviation. Connected components, defined as subgraphs with interconnected vertices, were identified to group features with high mutual similarity. Within each element, the feature with the highest degree of centrality was selected as the most representative.
2. **Spectral clustering** takes a global approach, embedding the feature graph into a lower-dimensional space while preserving its essential structural relationships. We computed the graph Laplacian L as: $L = D - A$, where D is the diagonal degree matrix, and A is the adjacency matrix. By analyzing eigenvectors corresponding to the smallest nonzero eigenvalues of the Laplacian, we obtained a low-dimensional representation of the feature graph. This approach naturally revealed clusters of related features. Features closest to cluster centroids in the embedded space are selected as representative features.

Figure 2 illustrates the workflow of our Graph-FS approach in radiomics. Supplementary Information provides further details about feature selection methods in this work.

Stability, consistency, and reproducibility evaluation metrics

In our analysis, three core concepts: stability, consistency, and reproducibility are essential for evaluating the reliability of radiomic biomarkers and ensuring their clinical utility across diverse settings.

1. *Stability*: Stability refers to the robustness of a feature selection method when subjected to minor variations in preprocessing parameters or data inputs[34]. In the context of radiomics, it measures how consistently a method selects features when the input data are perturbed, for example by changing image normalization scales or bin widths[34,35]. We quantified stability using multiple metrics:

   i. *Pearson's Correlation coefficient (PCC)*:

   PCC measures the correlation between the weights of selected feature subsets, providing a value between −1 and 1, where values closer to 1 indicate a strong positive correlation, 0 means no correlation, and −1 indicates a strong negative correlation. PCC is a symmetric stability measure, calculated as Eq. 1. Where μ_{xi} and μ_{yi} is the mean of feature set x and y, respectively.

$$PCC\left(x_i, y_j\right) = \frac{\sum \left(x_i - \mu_{xi}\right)\left(y_i - \mu_{yi}\right)}{\sqrt{\sum \left(x_i - \mu_{xi}\right)^2 \left(y_i - \mu_{yi}\right)^2}} \tag{1}$$

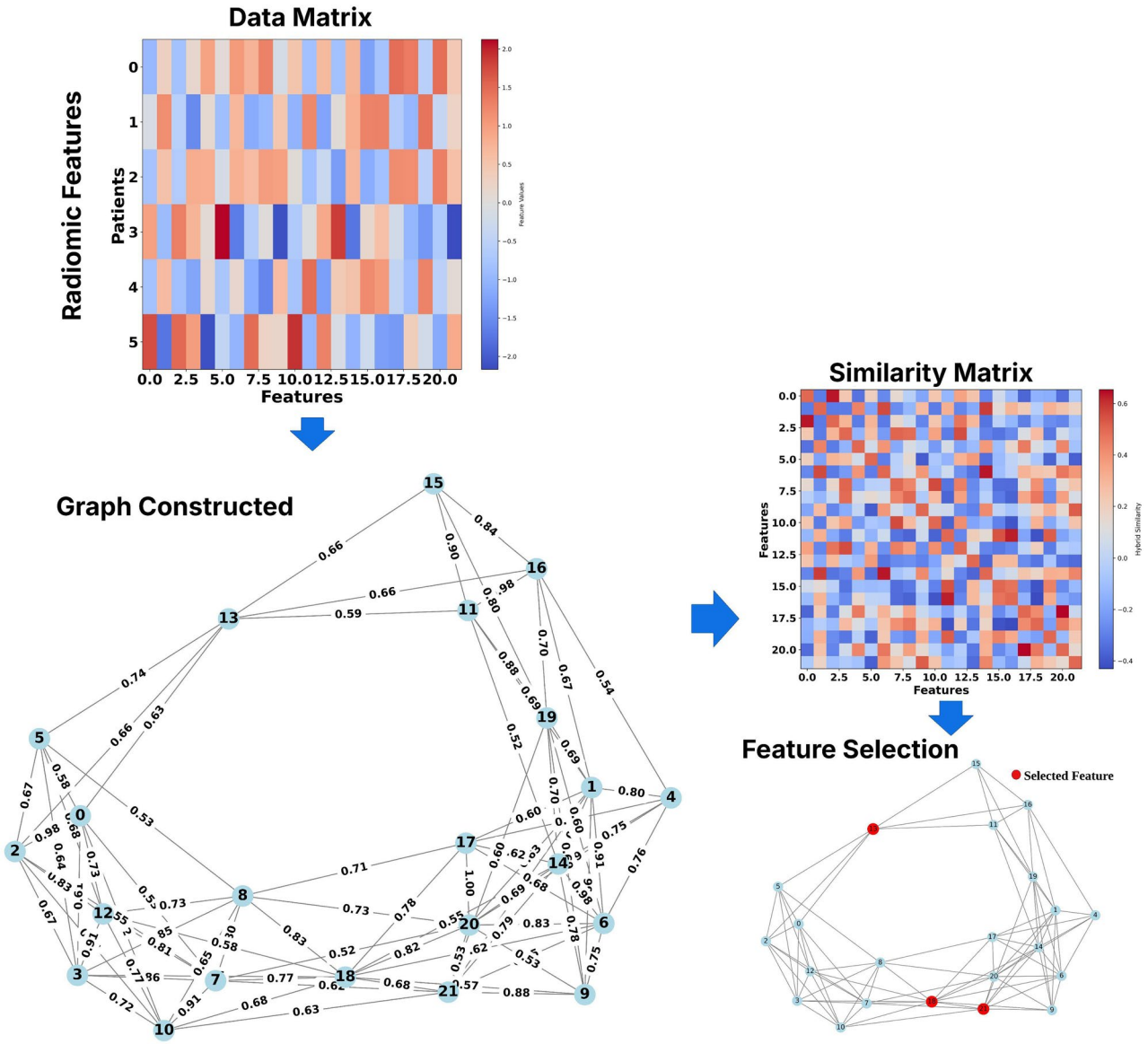

Fig. 2. Workflow of Graph-feature selection (Graph-FS) in radiomics. This figure illustrates the general workflow of Graph-FS applied to a sample radiomics feature data matrix with six patients (rows) and 22 features (columns). The radiomics feature matrix is normalized to reflect feature values across patients. A graph is constructed using a similarity matrix, which can be generated through various methods, such as Spearman correlation, cosine similarity, modular information, or Laplacian computation. Nodes represent features, and edges indicate their similarity. The resulting similarity matrix is used to identify clusters of related features, with the most representative features highlighted as red nodes.

ii. *Jaccard Index (Jaccard Similarity coefficient, JSC)*

The Jaccard Index quantified the similarity between pairs of selected feature subsets. It is calculated as shown in *Eq. 2, where x and y present two feature sets, and* $|x_i \cap y_j|$ represents the intersection size and $|x_i \cup y_j|$ the union size of sets x and y[36].

$$J(x_i, y_j) = \frac{|x_i \cap y_j|}{|x_i \cup y_j|} = \frac{|x_i \cap y_j|}{|x_i| + |y_j| - |x_i \cap y_j|} \tag{2}$$

Values range from 0 (no overlap) to 1 (identical sets). For overall stability, the Jaccard Index across pairs of selected feature subsets p is average:

$$JS = \frac{2}{|P| \times (|P-1|)} \sum_{i=1}^{|P|-1} \sum_{j=i+1}^{|P|} J(x_i, y_j) \tag{3}$$

iii. *Dice-Sorensen's Index (DSI)*[37] is also known as *Dice Similarity Coefficient (DSC)* quantifies similarity between two sets as represented in Eq. 4.

$$DSI\ (x_i, y_j) = \frac{2\,|x_i \cap y_j|}{|x_i| + |y_j|}. \tag{4}$$

iv. *Overlap Percentage* calculates the proportion of shared elements between two sets relative to the smaller set's size, as shown in Eq. 5.

$$overlap(x, y) = \frac{|x \cap y|}{\min(|x|, |y|)} \tag{5}$$

.

2. *Consistency* (Ranking) refers specifically to the stability of feature rankings, that is, how consistently a method ranks the importance of features across variations in settings or datasets. We evaluated consistency using Kendall's Coefficient of Concordance (W), which quantifies agreement among ranked feature lists. High Kendall's W values indicate strong agreement in feature importance rankings, reflecting the method's robustness in prioritizing the most relevant features across changing conditions.

   i. *Kendall's Coefficient of concordance(W)* Kendall's W is a nonparametric measure of agreement between rankings, used here to evaluate consistency in feature ranking across different radiomics settings[38].

Given n radiomics settings and m features, Kendall's W is calculated using Eq. 6:

$$W = \frac{12S}{m^2.(n^3 - n)} \tag{6}$$

where S is the sum of squared deviations of ranks, here, S is computed using Eq. 7.

$$S = \sum_{i=1}^{m} (R_i - \bar{R})^2 \tag{7}$$

3. *Reproducibility* measures the extent to which the selected features remain consistent across different institutions or datasets rather than within the same dataset under perturbation. This is critical for clinical translation, as reproducible biomarkers are more likely to generalize across imaging centers, scanners, or patient populations. We assessed reproducibility using the Jaccard Index to measure the overlap in feature sets selected independently from each center. The final reproducibility score was calculated as the average pairwise Jaccard Index across all combinations of centers.

This comprehensive assessment of stability, consistency, and reproducibility is crucial for developing radiomic biomarkers that are reliable and applicable in diverse clinical settings. The Supplementary Information provides detailed formulas for each metric.

Model training and evaluation for survival prediction

We implemented five machine learning classifiers Random Forest, XGBoost[39]CatBoost[40]AdaBoost[41]and LightGBM[42] to predict 2-year survival from radiomic features. These models were chosen because they effectively manage high-dimensional and multicollinear data, reduce noise, and adapt well to the imbalanced class distributions typical in survival analysis. Each classifier was trained using the top 20 most frequently selected features identified by each feature selection method.

For model development, we combined patient data from two institutions: the University of Maryland, Baltimore (UMB; Center 1) and MD Anderson Cancer Center (Center 2). We restricted model training to a homogeneous subgroup of patients receiving definitive chemoradiation therapy, which represented the largest consistent treatment group (401 patients out of 614 total from training centers 1 and 2).

Hyperparameters for each classifier were optimized using GridSearchCV with an internal 3-fold cross-validation strategy (see Supplementary Table S3 for detailed parameters). We trained the models using repeated 10-fold stratified cross-validation (three repeats) to ensure reliable performance estimates and minimize variance. This training cohort had a significant class imbalance, consisting of 350 survivors (87%) and 51 deceased patients (13%). To address this imbalance, Adaptive Synthetic Sampling (ADASYN) was applied exclusively to the training partitions within each iteration of the cross-validation, increasing the minority-to-majority class ratio from approximately 0.15 (51:350) to nearly 1:1 (353:350). This rigorous oversampling approach preserved the integrity of internal validation folds while ensuring stable and unbiased performance estimation.

Within each iteration, ADASYN oversampling was applied only to the training partition (9 out of 10 folds), preserving the integrity of the internal validation set. Model performance was averaged across 30 runs (10 folds × 3 repetitions).

Model generalizability was assessed using an independent external validation cohort from UMMC (Center 3), which included a broader and more heterogeneous patient population treated with various modalities (e.g., surgery alone, radiation alone). This design allowed us to evaluate the robustness and clinical relevance of our models across different clinical scenarios. Figure 3 summarizes the overall strategy for model training, internal validation, and external testing. The primary evaluation metric was the area under the receiver operating characteristic curve (ROC-AUC), which measures the model's ability to distinguish between survival outcomes by balancing sensitivity and specificity across varying thresholds. To provide more reliable estimates, we calculated the bootstrapped mean AUC with 95% confidence intervals (CIs) based on 1,000 bootstrap iterations. Additional metrics included the F1 score, assessing the balance between precision and recall, and accuracy, to report the overall proportion of correctly classified cases.

All analyses were performed on a Windows 11 Pro workstation equipped with an Intel Core i9-14900 K processor (32 cores) and an NVIDIA GeForce RTX 4070 GPU for accelerated computations. The primary evaluation metric for assessing model performance was the area under the receiver operating characteristic curve (ROC-AUC).

Software

All preprocessing steps were conducted in Python 3.11 using the latest versions of SimpleITK (v2.3.1, https://simpleitk.readthedocs.io/en/master/) to ensure high-quality image processing and precise isotropic resampling. We extracted a comprehensive range of radiomic features, including texture, shape, and first-order characteristics, from pre-treatment CT images using PyRadiomics (v3.1.0, https://pyradiomics.readthedocs.io/en/latest/). For feature selection, we utilized scikit-learn (v1.3.0, https://scikit-learn.org/stable/), mRMR-selection (v0.2.8, https://pypi.org/project/mrmr-selection/), and the networks package (v3.1) to implement methods such as Boruta, Lasso, RFE, mRMR, and Graph-FS. Feature stability and reproducibility were assessed using the Jaccard Index and DSI calculated via Python's set operations. Kendall's W, a measure of concordance, was computed using statsmodels (v0.14.0, https://www.statsmodels.org/stable/index.html) and pingouin (v0.5.3, https://pingouin-stats.org/). For classification, we employed several ensemble classifiers, including:

- XGBoost (Extreme Gradient Boosting, v3.0.2, https://xgboost.readthedocs.io/en/stable/),
- CatBoost (Categorical Boosting, v1.2.8, https://catboost.ai/), and.
- LightGBM (Light Gradient Boosting Machine, v4.6.0, https://lightgbm.readthedocs.io/en/latest/), implemented through scikit-learn's ensemble module and their respective Python APIs.

Results

Feature selection stability, consistency, and reproducibility

We evaluated the robustness of feature selection methods across varying radiomics settings using five complementary metrics (Fig. 4). These included the Pearson Correlation Coefficient (PCC; Fig. 4A), which measures the correlation of feature sets across parameter variations, with higher values indicating greater stability; Kendall's Concordance Coefficient (W; Fig. 4B), which assesses the consistency in feature rankings under different settings; the Jaccard Index (Fig. 4C) and Dice–Sørensen Index (Fig. 4D), both quantifying the

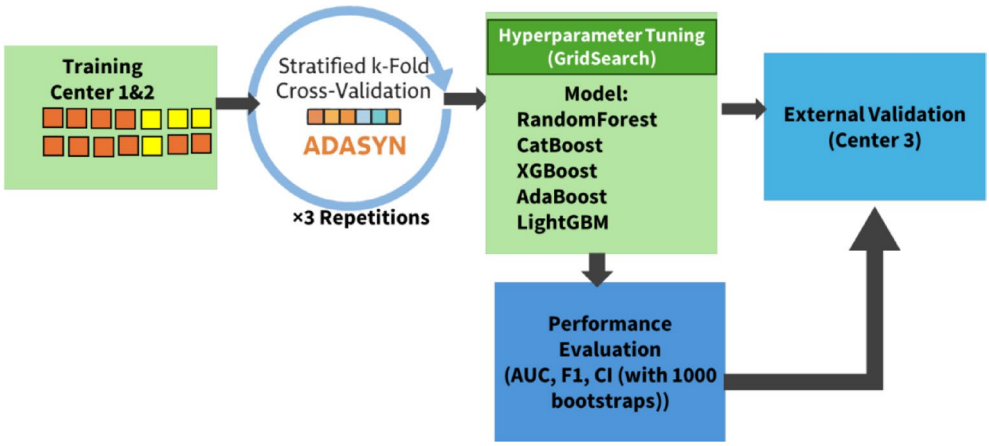

Fig. 3. Model training and validation strategy. Radiomic models were trained using stratified 10-fold cross-validation, repeated three times. In each split, ADASYN was applied to the training folds, keeping validation folds untouched. Hyperparameters were optimized using GridSearch. Five ensemble classifiers (Random Forest, CatBoost, XGBoost, AdaBoost, and LightGBM) were evaluated. The final performance was averaged over 30 runs and assessed using AUC, F1 score, and 95% confidence intervals (via 1000 bootstraps). External validation was performed on data from Center 3 to test generalizability.

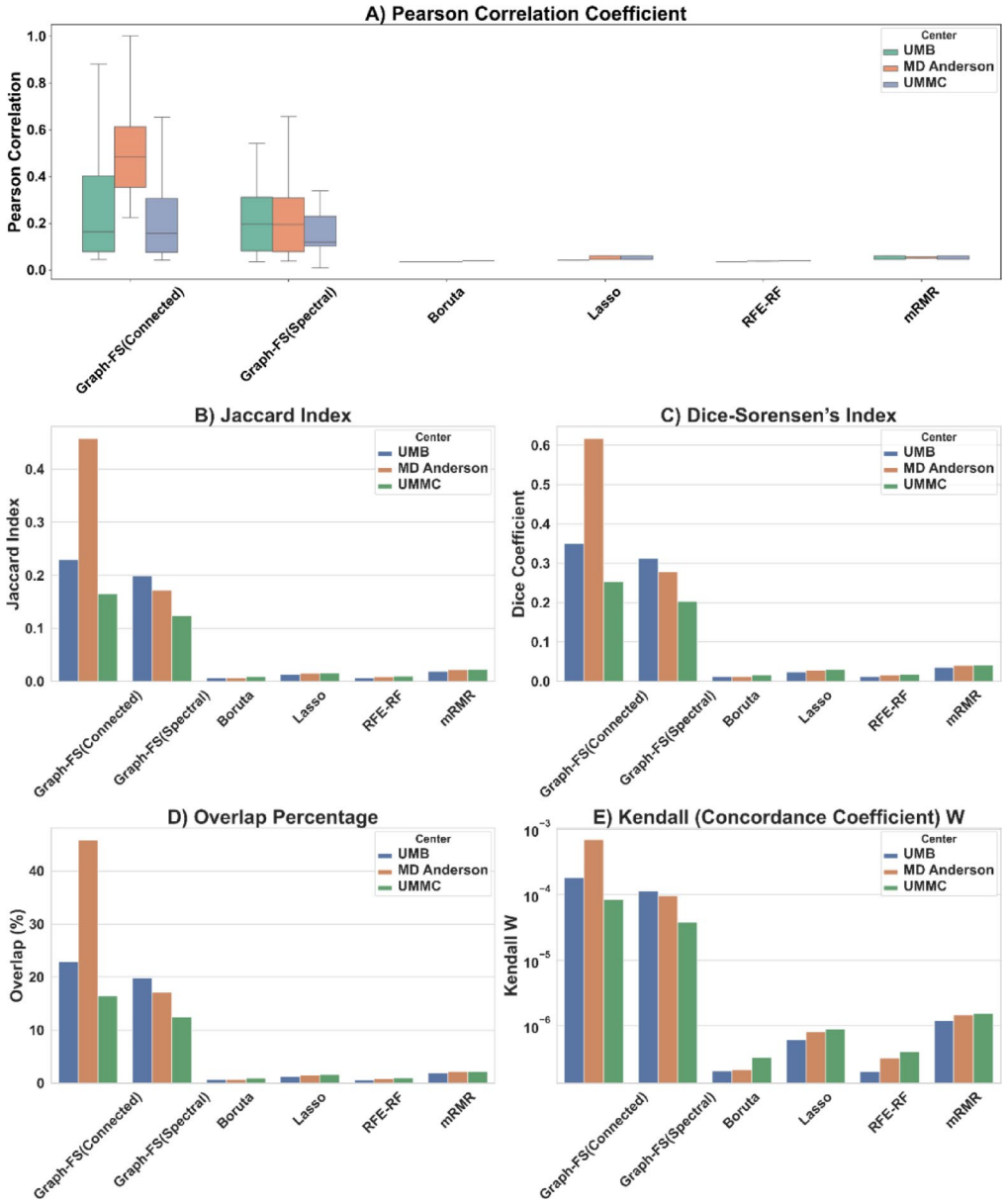

Fig. 4. Intra-Center stability and consistency of feature selection methods across radiomics parameter settings. This multi-panel figure compares the performance of feature selection methods across 36 radiomics parameter settings within three institutions: the University of Maryland, Baltimore (UMB), The University of Texas MD Anderson Cancer Center (MD Anderson), and the University MAASTRO Medical Center (UMMC). (**a**) Pearson Correlation Coefficient (PCC) quantifies the consistency of features selected under varying radiomics settings. Higher PCC values indicate greater consistency of feature selection across varying parameter settings. (**b**) the Jaccard Index measures the similarity between feature sets across different settings; (**c**) Dice-Sorensen's Index reflects the degree of overlap in selected features. (**d**) Overlap Percentage visualizing the proportion of shared features across radiomics configurations. (**e**) Kendall's Concordance Index (W) indicates feature ranking consistency across parameter settings for each method. Graph-FS (Connected) = Graph-Based Feature Selection (Connected Components), Graph-FS (Spectral) = Graph-Based Feature Selection (Spectral Clustering), Lasso = Least Absolute Shrinkage and Selection Operator, mRMR = Minimum Redundancy Maximum Relevance, RFE-RF = Recursive Feature Elimination with Random Forest.

similarity and overlap between selected feature sets; and the Overlap Percentage (Fig. 4E), which represents the proportion of consistently selected features across multiple radiomics parameter configurations.

Cross-center reproducibility is illustrated in Fig. 5, highlighting radiomic features that were commonly selected across Centers 1, 2, and 3.

The top ten most frequently selected radiomic features for each method are displayed in Fig. 6. The top 20 most frequently selected radiomic features using Graph-FS (Connected Components) for each center are shown in Figures S1, S2, and S3.

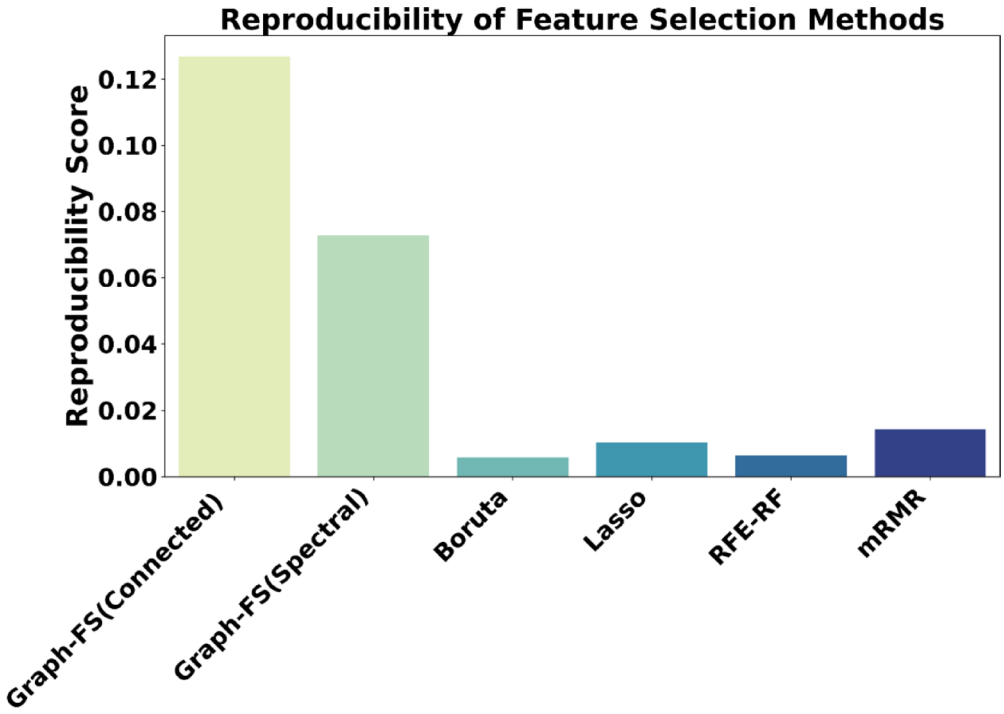

Fig. 5. Bar plot of feature selection reproducibility scores across multiple centers. This bar plot illustrates the reproducibility scores of various feature selection methods across different medical centers. Higher scores reflect more consistent feature selection across centers, indicating higher reproducibility. Graph-FS (Connected) = Graph-Based Feature Selection (Connected Components), Graph-FS (Spectral) = Graph-Based Feature Selection (Spectral Clustering), Lasso = Least Absolute Shrinkage and Selection Operator, mRMR = Minimum Redundancy Maximum Relevance, RFE-RF = Recursive Feature Elimination with Random Forest.

Survival prediction performance

Table 2 presents the classification performance of CatBoost, Random Forest, and XGBoost for survival prediction. Evaluation metrics include the area under the ROC curve (ROC-AUC) with 95% confidence intervals, bootstrapped AUC, F1 score, and accuracy. Additional performance results for AdaBoost and LightGBM are provided in Supplementary Tables S4 and S5.

Figure 7 displays the ROC curves for the external validation cohort (Center 3) across all feature selection methods and five classifiers, highlighting the generalizability of the models to heterogeneous clinical populations. For comparison, the ROC curves obtained from the training set are shown in Supplementary Figure S4.

Discussion

The reproducibility of scientific findings is not a cornerstone of research but a critical barrier to clinical translation, particularly in radiomics. Variability in datasets, feature extraction methodologies, and parameter settings significantly impacts the reliability of selected features, ultimately limiting their translational potential. This research directly addresses this challenge by demonstrating the effectiveness of Graph-FS in identifying stable and reproducible features, an essential step toward developing robust AI-driven models in multi-institutional clinical settings.

Reproducibility and stability across centers

Graph-FS (Connected Components) achieved the highest Dice Similarity Index (DSI = 0.62) for stability within centers, significantly outperforming Graph-FS (Spectral Clustering, DSI = 0.28), and conventional methods such as Boruta (0.01), Lasso (0.03), mRMR (0.04), and RFE-RF (0.03).

For cross-center reproducibility, Graph-FS (Connected Components) achieved a Jaccard Index of 0.127, compared to Boruta (0.005), Lasso (0.010), RFE-RF (0.006), and mRMR (0.014).

Additionally, Graph-FS demonstrated superior ranking consistency, achieving the highest Kendall's W (6.9×10^{-3}), outperforming Boruta (2.1×10^{-7}), Lasso (8.0×10^{-7}), mRMR (1.4×10^{-6}), and RFE-RF (3.18×10^{-7}).

Although Kendall's W values were low across all methods, indicating inherent instability in feature ranking under varying parameter settings, Graph-FS exhibited a comparative advantage in maintaining selection consistency.

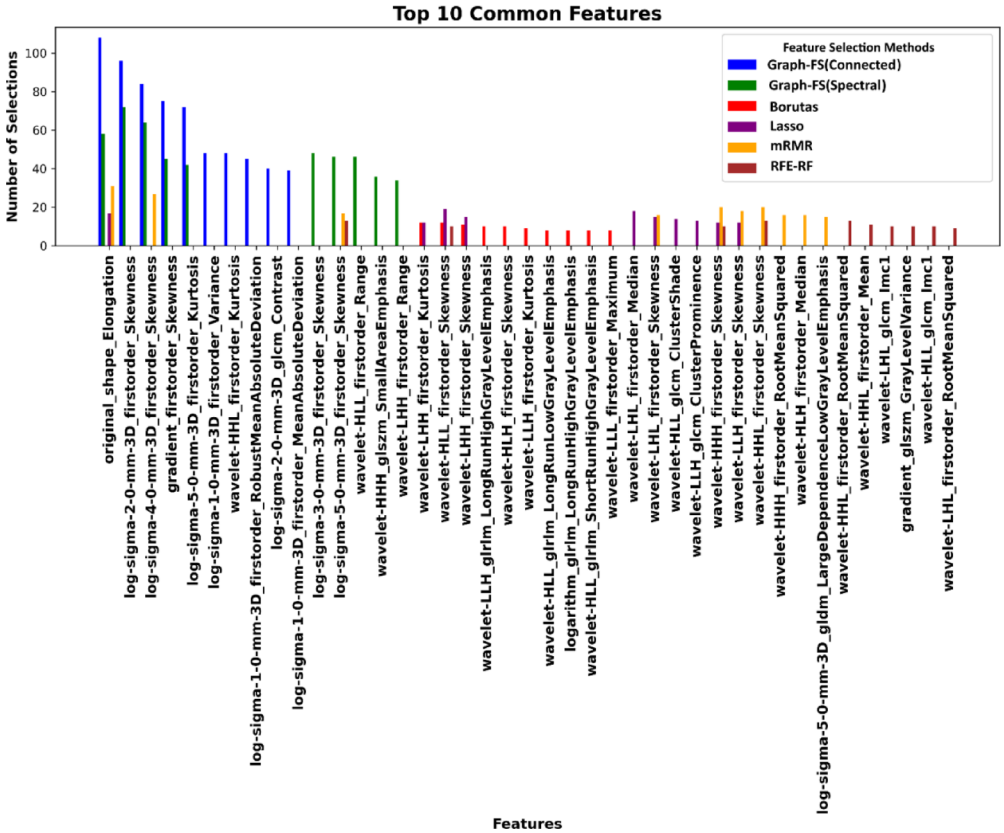

Fig. 6. Frequently selected radiomic features across feature selection methods. This bar chart presents the top 10 most frequently selected radiomic features identified by six different feature selection methods: Graph-FS (Connected Components), Graph-FS (Spectral Clustering), Boruta, Lasso, mRMR, and RFE-RF. The selection frequency of each feature is aggregated across 36 radiomic settings and three medical centers, reflecting the stability and reproducibility of each method. Graph-FS (Connected) = Graph-Based Feature Selection via Connected Components. Graph-FS (Spectral) = Graph-Based Feature Selection via Spectral Clustering. Lasso = Least Absolute Shrinkage and Selection Operator. mRMR = Minimum Redundancy Maximum Relevance. RFE-RF = Recursive Feature Elimination with Random Forest.

To summarize these improvements, we calculated the mean relative increase of Graph-FS (Connected Components) over each baseline method for the three reproducibility metrics, using the formula: $\frac{(\text{Graph-FS} - \text{Baseline})}{\text{Baseline}} \times 100\%$, averaged across comparisons.

Based on this approach, Graph-FS (Connected Components) showed an average improvement of 98.26% in feature stability (DSI), 98.10% in cross-center reproducibility (Jaccard Index), and 99.99% in ranking consistency (Kendall's W).

These results underscore the importance of prioritizing stable features across diverse conditions, rather than relying solely on rank-based importance features, which can be highly sensitive to parameter fluctuations.

Biological relevance of common radiomic features

Several of the most frequently selected features by Graph-FS (Connected Components), as highlighted in Fig. 5, demonstrate biological plausibility and known clinical relevance in head and neck cancer[2].

For example, original_shape_Elongation, quantifies the deviation of a tumor's shape from a perfect sphere, thereby measuring its elongation. Tumors with higher elongation values may reflect more invasive growth patterns, which are often associated with increased aggressiveness and poorer prognosis.

Intensity-based features such as log-sigma-2-0-mm-3D_firstorder_Skewness and wavelet-HHL_firstorder_Kurtosis capture the distributional asymmetry and peakedness of voxel intensities, respectively. These metrics reflect intratumoral heterogeneity, which is frequently associated with treatment resistance, hypoxia, and necrosis[43,44].

Additionally, texture features like SmallAreaEmphasis and ShortRunLowGrayLevelEmphasis, which quantify fine-grained spatial variations in gray-level distributions, may capture biologically relevant patterns such as tumor compactness, fibrosis, or fragmented necrotic regions, factors known to influence both radiation response and clinical outcomes[45,46].

Feature selection methods	Classifier	AUC (95% CI)	AUC Bootstrapped mean	F1 score	Accuracy
Graph-FS (connected component)	CatBoost	0.70(0.59–0.8)	0.70	0.795	0.69
	RandomForest	0.67(0.56–0.77)	0.67	0.82	0.72
	XGBoost	0.65(0.53–0.76)	0.65	0.81	0.72
Graph-FS (spectral clustering)	CatBoost	0.61 (0.5–0.72)	0.61	0.67	0.57
	RandomForest	0.55(0.44–0.68)	0.55	0.68	0.56
	XGBoost	0.57 (0.462–0.69)	0.57	0.70	0.59
Boruta	CatBoost	0.61 (0.46–0.76)	0.61	0.53	0.47
	RandomForest	0.64(0.51–0.76)	0.64	0.5	0.46
	XGBoost	0.62 (0.48–0.74)	0.62	0.20	0.29
Lasso	CatBoost	0.53 (0.38–0.68)	0.53	0.80	0.70
	RandomForest	0.58 (0.45–0.71)	0.56	0.86	0.77
	XGBoost	0.58 (0.42–0.71)	0.57	0.84	0.74
mRMR	CatBoost	0.50 (0.37–0.62)	0.50	0.78	0.66
	RandomForest	0.56(0.44–0.68)	0.57	0.78	0.65
	XGBoost	0.54 (0.40–0.66)	0.54	0.78	0.66
RFE-RF	CatBoost	0.49 (0.36–0.61)	0.48	0.84	0.74
	RandomForest	0.43(0.31–0.55)	0.43	0.86	0.75
	XGBoost	0.46 (0.34–0.58)	0.46	0.81	0.68

Table 2. Performance of feature selection methods in 2-year survival predictions with CatBoost, Random Forest, and XGBoost classifier on the external testing set. Graph-FS = Graph-Based Feature Selection, Lasso = Least Absolute Shrinkage and Selection Operator, mRMR = Minimum Redundancy Maximum Relevance, RFE-RF = Recursive Feature Elimination with Random Forest.

Clinical implications

Graph-FS (Connected Components) achieved the highest AUC during external validation using CatBoost (AUC = 0.70), outperforming Graph-FS (Spectral Clustering, AUC = 0.61), Boruta (AUC = 0.61), mRMR (AUC = 0.57), Lasso (AUC = 0.53), and RFE-RF (AUC = 0.48) (Table 2).

While traditional methods such as Boruta and Lasso performed well during training (Figure S1), their AUCs declined significantly on the external test set. This performance drop suggests overfitting, where models may have captured patterns specific to the training data rather than learning generalizable signals.

In contrast, Graph-FS (Connected Components) maintained consistent performance across both internal and external datasets, underscoring its robustness. This stability suggests that the method selects features that are both reproducible and generalizable across diverse patient populations and imaging protocols. Unlike approaches that rely solely on feature rankings or importance scores, Graph-FS leverages graph structures to identify topologically important features, capturing deeper and more biologically meaningful relationships among radiomic variables.

These findings underscore the clinical value of Graph-FS. Its consistent cross-center performance, along with enhanced feature stability and reproducibility (Figs. 4, 5 and 6), supports its potential as a reliable, noninvasive tool for identifying imaging biomarkers predictive of clinical outcomes in head and neck cancer.

Comparison with prior studies

Previous work suggests that methods like Lasso and mRMR balance stability and predictive performance in radiomics, particularly due to their simplicity and reduced hyperparameter tuning requirements[47]. Our findings align with these observations, with Lasso and mRMR performing better than conventional methods. However, the superior performance of Graph-FS reinforces the advantage of graph-based approaches in capturing complex feature relationships and mitigating variability, further supporting its clinical potential.

Limitations and future directions

While our findings demonstrate that Graph-FS significantly improves feature stability and reproducibility, this evaluation was limited to head and neck cancer using CT imaging. To assess its broader applicability, future work should extend this framework to other tumor types and imaging modalities, such as MRI and PET.

Although we propose potential biological relevance for the most frequently selected features based on prior literature, we did not directly validate these associations using histopathological or genomic data. Future studies that integrate radiomics with molecular or pathological correlations are needed to understand the biological mechanisms behind these imaging biomarkers.

Graph-based feature selection is an area of growing interest, with emerging techniques like adaptive graph-based selection, node dynamic mechanisms, and deep learning approaches showing significant potential[48,48,49,51]. Future research should explore how combining these techniques with Graph-FS can enhance performance further.

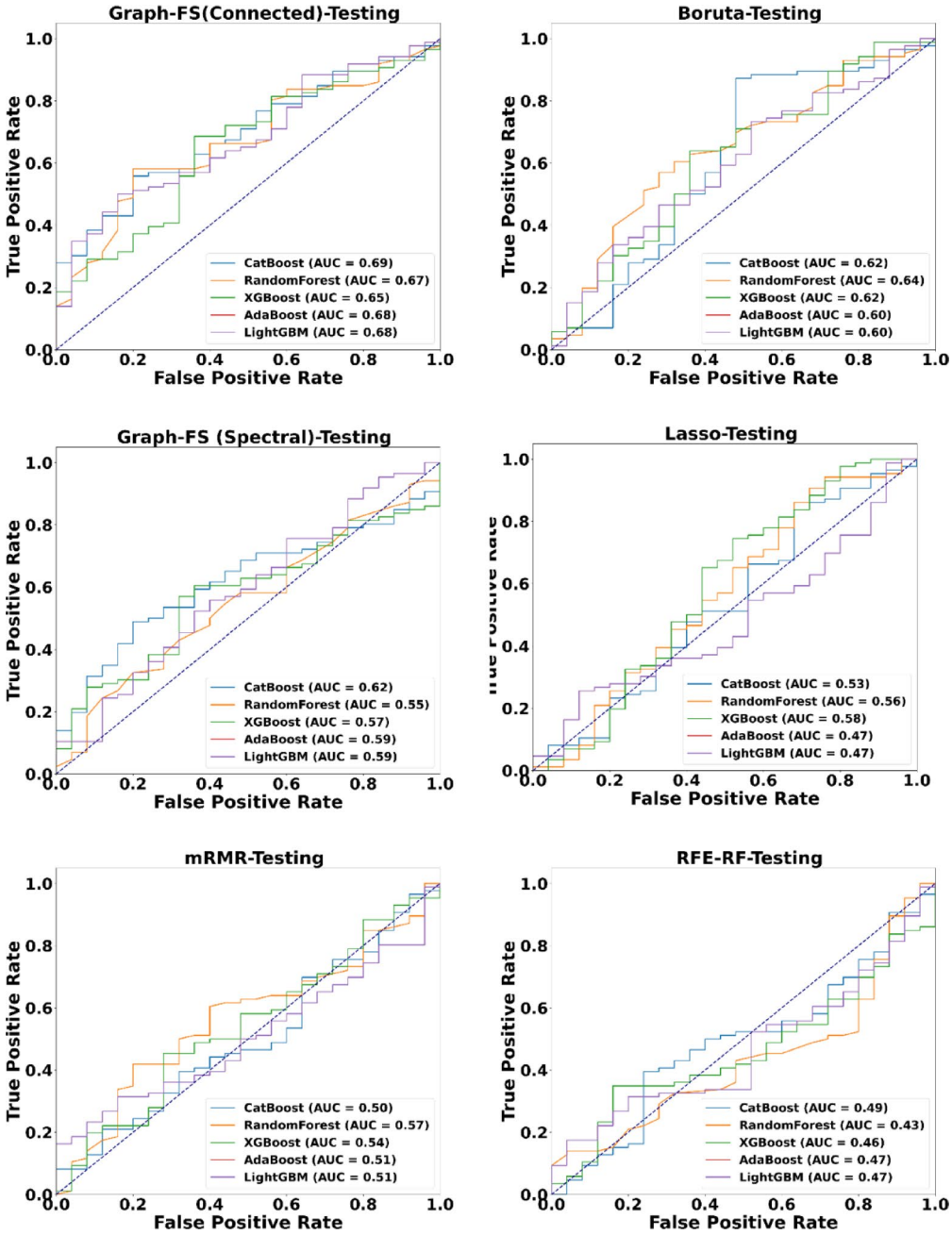

Fig. 7. Receiver Operating Characteristic (ROC) curves for 2-year survival prediction on the external validation cohort (UMMC, Center 3). Six feature selection methods—Graph-FS (Connected Components), Graph-FS (Spectral Clustering), Boruta, Lasso, mRMR, and RFE-RF—were evaluated using five classifiers: CatBoost, Random Forest, XGBoost, AdaBoost, and LightGBM. Each subplot displays ROC curves for a given method, with AUC values shown in the legend. These results reflect the generalizability and predictive performance of each method-classifier combination when applied to an independent, heterogeneous patient cohort. Graph-FS (Connected) = Graph-Based Feature Selection using Connected Components; Graph-FS (Spectral) = Graph-Based Feature Selection using Spectral Clustering; Lasso = Least Absolute Shrinkage and Selection Operator; mRMR = Minimum Redundancy Maximum Relevance; RFE-RF = Recursive Feature Elimination with Random Forest.

Conclusion

Traditional feature selection methods, such as Boruta, Lasso, and Recursive Feature Elimination (RFE), are highly dependent on outcome labels and sensitive to parameter changes, often leading to inconsistent feature selection. In contrast, Graph-Feature Selection (Graph-FS) provides a label-independent framework that enhances reproducibility by consistently identifying stable features across diverse radiomic settings. This study

demonstrates that Graph-FS offers a robust solution to a fundamental challenge in radiomics: the variability of traditional feature selection methods.

Beyond radiomics, Graph-FS has the potential to support AI-driven predictive modeling and large-scale biomarker discovery, contributing to the advancement of precision medicine. Its adaptability across datasets and compatibility with deep learning frameworks make it a valuable tool for integrating radiomics into clinical workflows. By improving feature selection stability, Graph-FS facilitates personalized treatment strategies and accelerates the clinical adoption of radiomics-driven decision-making in oncology.

Data availability

The datasets analyzed during the current study are not publicly available due to institutional data use agreements and patient privacy restrictions. However, de-identified data are available from the corresponding author on reasonable request. All data were anonymized in compliance with HIPAA regulations.The code used in this study is available on GitHub at https://github.com/hmMed22/GFSIR.1. However, certain components are subject to institutional licensing agreements and may be restricted. Full access may be granted upon publication and upon reasonable requests with the necessary permission.

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Author contributions

H.M. and L.R. conceptualized the study. Data curation was performed by H.M., X. L., J.M., L.S., E. A., H.T., D.A., M.F., R. K., W.S.M., P.T., A.S., D.A.G., and L.R. H.M. conducted the formal analysis. Funding acquisition was managed by L.R. and D.A.G. The investigation was carried out by H.M. and L.R. Project administration and software development were handled by H.M. under the supervision of L.R. H.M. wrote the original draft and all authors reviewed and approved the final manuscript.

Declarations

Ethics declaration

This study was conducted following the ethical standards of the Institutional Review Board (IRB) at all participating institutions and with the principles outlined in the Declaration of Helsinki. An IRB-approved waiver of informed consent was granted, as this retrospective study involved de-identified patient data and did not require individual consent. All patient data were anonymized to protect confidentiality, in full compliance with HIPAA regulations for U.S.-based research. Additionally, the study was registered under Clinical Trials Registry Number HP-00101754.

Competing interests

The authors declare no competing interests.

Additional information

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